

Exercises

CHAPTER (1/2/3/4/5)

1. Enhance the `hello_with_time.php` script to display the current date as well as the time. Comment your code for readability. (Hint: With the `date()` function, you can use `M` to display the month name, `j` to display the day of the month, and `Y` to display the year.)
2. Write a script that creates a variable and assigns an integer value to it, then adds 1 to the variable's value three times, using a different operator each time. Display the final result to the user.
3. Write a script that creates two variables and assigns a different integer value to each variable. Now make your script test whether the first value is
 - a. Equal to the second value
 - b. greater than the second value
 - c. less than or equal to the second value
 - d. not equal to the second value and output the result of each test to the user.
4. Write a script that counts from 1 to 10 in steps of 1. For each number, display whether that number is an odd or even number, and also display a message if the number is a prime number. Display this information within an HTML table.
5. Modify the homing pigeon simulator to simulate two different pigeons on the same map, both flying to the same home point. The simulation ends when both pigeons have arrived home.
6. Using the `printf()` function, write a single line of code that takes a month (from 1 to 12), a day (from 1 to 31), and a four - digit year, and displays the resulting date, formatted in `mm/dd/yyyy` format. Don ' t forget to add a zero in front of the month or day if it's less than 10.
7. Write a script that emulates the function call `str_pad( $myString, $desiredLength)`. In other words, take a string, and add space characters to the right of it until the string reaches the desired length. Display both the original and padded string in the page.

CHAPTER (6/7)

1. Imagine that two arrays containing book and author information have been pulled from a database:

```
$authors = array( "Steinbeck", "Kafka", "Tolkien", "Dickens", "Milton", "Orwell" );  
$books = array(  
    array("title" => "The Hobbit", "authorId" => 2, "pubYear" => 1937),  
    array("title" => "The Grapes of Wrath", "authorId" => 0, "pubYear" => 1939),
```

```

array("title" => "A Tale of Two Cities", "authorId" => 3, "pubYear" => 1859),
array("title" => "Paradise Lost", "authorId" => 4, "pubYear" => 1667),
array("title" => "Animal Farm", "authorId" => 5, "pubYear" => 1945),
array("title" => "The Trial", "authorId" => 1, "pubYear" => 1925));

```

Instead of containing author names as strings, the \$booksarray contains numeric indices (keyed on "authorId") pointing to the respective elements of the \$authors array. Write a script to add an "authorName" element to each associative array within the \$books array that contains the author name string pulled from the \$authors array. Display the resulting \$books array in a Web page.

2. Imagine you are writing a version of the computer game Minesweeper. Use arrays to create and store a minefield on a 20 x 20 grid. Place ten mines randomly on the grid, then display the grid, using asterisks (*) for the mines and periods (.) for the empty squares. (Hint: To return a random number between 0 and 19 inclusive, use rand(0, 19).)
3. Write a function that takes an array argument, and returns a string containing XHTML markup for a definition list where each key of the array is a term, and each corresponding value is a definition.

(Hint: An XHTML definition list element consists of < dl > ... </dl > tags. Inside these tags, terms are marked up using < dt > ... </dt > tags, and definitions using < dd > ... </dd > tags.)

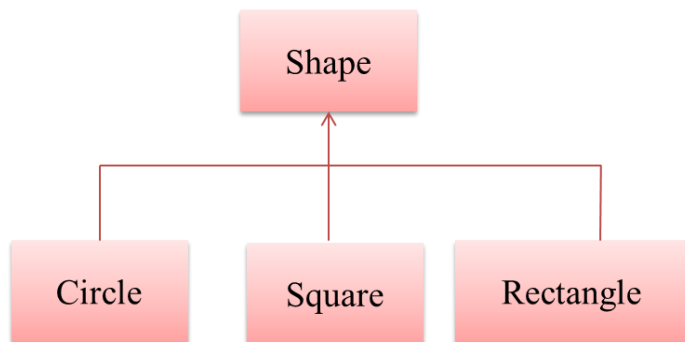
4. A factorial of any given integer, n, is the product of all positive integers between 1 and nine inclusive. So the factorial of 4 is $1 \times 2 \times 3 \times 4 = 24$, and the factorial of 5 is $1 \times 2 \times 3 \times 4 \times 5 = 120$. This can be expressed recursively as follows:

- ❑ If $n == 0$, return 1. (This is the base case)
- ❑ If $n > 0$, compute the factorial of $n - 1$, multiply it by n, and return the result

Write a PHP script that uses a recursive function to display the factorials of the integers 0 to 10.

CHAPTER (8)

1. Write program that you create Circle, Square and Rectangle class and then these classes are derived from the Shape class and should be added getArea() method of each class. (refer to Try It Out exercise). **[Inheritance – Exercise]**



Creating Shape Classes using Inheritance

My Circle

My circle has a radius of 4.

It is red and it is filled.

The area of my circle is: 50.265482457437.

My Square

My square has a side length of 3.

It is green and it is hollow.

The area of my square is: 9.

My Square

My square has a side length of 5 and 3

It is yellow and it is hollow.

The area of my square is: 15.

- Write program that you create Circle and Square class and then these classes are derived from the Shape class and should be added getArea() method of each class. (refer to Rectangle class) **[Abstract Class – Exercise]**

Circle Area= $M_PI * \text{pow}(\$this->_radius, 2)$

Square Area= $\text{pow}(\$this->_sideLength, 2)$

- Write a Calculator class that can store two values, then add them, subtract them, multiply them together, or divide them on request. For example:

```
$calc = new Calculator( 3, 4 );
```

```
echo $calc-> add(); // Displays "7"
```

```
echo $calc-> multiply(); // Displays "12"
```

- Create another class, CalcAdvanced, that extends (inherits from) the Calculator class. CalcAdvanced should be capable of storing either one or two values:

```
$ca = new CalcAdvanced( 3 );
```

```
$ca = new CalcAdvanced( 3, 4 );
```

CalcAdvanced should also add the following methods:

pow() that returns the result of raising the first number (the base) to the power of the second number

sqrt() that returns the square root of the first number

exp() that returns raised to the power of the first number

(Hint: PHP contains built - in functions called pow() , sqrt() , and exp().)

CHAPTER (9)

1. In previous Try It Out, you created three-step registration form using hidden form fields. Rewrite this script to input the entered form data for each three step.

The image displays four screenshots of a web application running on localhost, showing the progression of a three-step registration form.

- Step 1: Student Login**
The form is titled "Student Login : Step 1". It contains two input fields: "User Name" with the value "John" and "Password" with masked characters "*****". A "Next >" button is at the bottom right.
- Step 2: Course Form**
The form is titled "Course Form: Step 2". It includes radio buttons for "Major Faculty" (Software is selected), checkboxes for "Interesting Subject?" (Cloud Computing and Programming are selected), and a dropdown menu for "Select Your Year?" (Third Year is selected). "Back" and "Next >" buttons are at the bottom.
- Step 3: Course Form**
The form is titled "Course Form: Step 3". It includes a checkbox for "Do you want to receive our newsletter?" (checked), a text area for "Any comments?" with the text "Interest it!", and "Back" and "Next >" buttons at the bottom.
- Thank You**
The final screen displays the user's information: "Name: John" and "From Year: thirdYear". It also shows the selected subjects: "I am interesting Cloud Computing, Programming" and a confirmation message: "Thank you, your application has been received."

2. Write a simple number - guessing game in PHP. The script should “ think ” of a random number between 1 and 100, then give the user five chances to guess the number. For each guess, the script should report whether the guessed number was too low, too high, or correct. (Hint: Use rand(1, 100)to generate a random number between 1 and 100.)
3. Create a script that displays a form allowing the user to select one of three Amazon stores — amazon.com , amazon.ca, and amazon.co.uk — and then jumps to the relevant store based on the user ’ s choice.

CHAPTER (10)

1. Write a script that uses cookies to remember how long ago a visitor first visited the page. Display this value in the page, in minutes and seconds.
2. In Chapter 9 you created a three - step registration form using hidden form fields. Rewrite this script to use sessions to store the entered form data, so users can come back to the form at another time and continue where they left off. Remember to erase the entered data from the session once the registration process has been completed.

CHAPTER (12)

1. Write out an SQL statement that creates a table called members in your mydatabase database to store information about the members of a book club. Store the following data for each person: first name, last name, age, and the date they joined the club. Create more SQL statements to insert five imaginary people into this table:
 - ☐ Jo Scrivener, aged 31, joined September 3, 2006
 - ☐ Marty Pareene, aged 19, joined January 7, 2007
 - ☐ Nick Blakeley, aged 23, joined August 19, 2007
 - ☐ Bill Swan, aged 20, joined June 11, 2007
 - ☐ Jane Field, aged 36, joined March 3, 2006
2. Write a PHP script to query the table you created in Exercise 1, displaying the details of all club members under 25 years of age.